Program Electives and Specializations for JRB MTech Robotics

AU-wise PE list

BIRD

JRD8990 MTP-II (0-0-24)
JRL8880 Special Topics in Robotics I (3-0-0)
JRL8881 Special Topics in Robotics II (3-0-0)
JRV8880 Special Module in Robotics I (1-0-0)
JRS8880 Independent Study (0-3-0)
JRL8730 Aerial Robotics (3-0-0)

EE

ELL8299	Large language models
ELL8298	Cognitive Architectures for Agentic AI: Building Intelligent Autonomous Systems
ELL8284	Natural Language Processing
ELL8122	Model Predictive Control
ELL8120	Model Reduction for Control
ELL8102	Optimal Control
ELL8101	Learning-Based Control
ELL8100	Nonlinear Control Systems
ELL7325	Introduction to sensors technology
ELL7283	Embedded Systems
ELL7125	Control of Robotic Systems
ELL8196	Selected Topics in Control
ELL8123	Adaptive Control
ELL7122	Control of Networked & Complex Systems

ScAI

AIL7002	Data Structures and Algorithms
AIL7011	Numerical Optimization
AIL7014	Introduction to Optimization
AIL7021	Deep Learning
AIL7022	Reinforcement Learning
AIL7023	Graph Machine Learning
AIL7031	Deep Learning for Computer Vision

AIL7051	Bayesian Statistical Modelling and Computation
AIL7061	Computational Cognition
AIL7062	AI for Computational Biology
AIL8001	Mathematics of Machine Learning
AIL8024	Computational Learning Theory
AIL8027	Advanced Reinforcement Learning
AIL8062	Special Topics in Computer Vision
AIL8023	Special Topics in Reinforcement Learning

CS

COL7375	Deep Learning
COL7560	Machine Learning for Networked Systems
COL7151	Algorithmic graph theory
COL7155	Algorithmic Game Theory
COL7154	Approximation Algorithms
COL7153	Complexity theory
COL7156	Mathematical Programming
COL7681	Computer Graphics
COL7683	Digital Image Analysis
COL7322	Introduction to Compressed Sensing
COL7372	Natural Language Processing
COL7378	Principles of Autonomous Systems
COL8628	Advanced Computer Vision

ME

MEL7103	Operations Planning and Control
MEL7101	Mathematical Programming
MEL7305	Computational Methods
MEL7234	Principles of Human Movement
MEL7216	Machine Learning for Computational Design
MEL7118	Reinforcement Learning for Operations Research
MEL7116	Optimization
MEL7115	Network Models for Public Systems
MEL7307	Smart Manufacturing
MEL7225	Design Principles for Precision

AM

AML7800	Deep Learning for Mechanics
AML7810	Probabilistic Machine Learning for Mechanics

AML7820 Uncertainty quantification and propagation

CART

CTL7003 Introduction to Electric Vehicles
CTL7008 Design and Control of EV Powertrain
CTL7005 Engineering of electric vehicles
CTL7032 Vehicle propulsion and transmissions
CTL7049 Digital Control Design for EV applications

SeNSE

DSL7711 Sensors and transducers

CBME

BML8800 Healthcare Wearables: Design and Applications
BML7700 Fundamentals of Biomechanics

Math

MTL7012 Computational Methods for Differential Equations
MTL7018 Computational Geometry
MTL7237 Introduction to Differential Geometry
MTL7763 Introduction to Game Theory
MTL7777 Introduction to Statistical Learning
MTL7799 Mathematical Analysis in Learning Theory
MTL7890 Optimization Algorithms
MTL8532 Bayesian Statistics and Modelling
MTL7742 Analysis and Design of Algorithms

JRB Specializations

Students may be awarded a specialization upon satisfying the following requirements:

1. Completion of at least 6 PE/OC credits (typically two courses) from the approved course list of the specialization.
2. Completion of both MTP-I and MTP-II in the chosen specialization.
3. Submission of an application to the PEC, specifying **one** specialization and demonstrating that the prescribed coursework and MTP requirements have been satisfied.

Based on the student's coursework and MTP, the PEC shall recommend the specialization to be awarded at the time of graduation. The PEC's decision shall be final.

The PEC may revise the approved course lists from time to time in response to curriculum revisions, course availability, or the introduction of new courses.

1. **Collaborative Robotics**
   - At most one course in total from the Advanced Control or Autonomy basket
   - ELL7122 Control of Networked & Complex Systems
   - AIL8027 Advanced Reinforcement Learning
   - MEL7115 Network Models for Public Systems
   - MTL7763 Introduction to Game Theory/ COL7155 Algorithmic Game Theory
   - COL7151 Algorithmic graph theory
   - AIL7023 Graph Machine Learning/COL7560 Machine Learning for Networked Systems
   - CTL7013 Connected and Autonomous Vehicles
   - COL7655 Foundations of Visual Computing

2. **Soft and Bio-inspired Robotics**
   - At most one course in total from the Advanced Control or Computer Vision basket
   - AML7630 Soft Robotics
   - BML7700 Fundamentals of Biomechanics/MEL7234 Principles of Human Movement/AML7620 Advanced Biomechanics/BML8700 Mechanics of Biological Systems
   - SBV7050 Bioinspiration and Biomimetics
   - JRL7999 Wearable Robotics
   - BML8800 Healthcare Wearables: Design and Applications
   - AML7800 Deep Learning for Mechanics/AML7810 Probabilistic Machine Learning for Mechanics/MEL7216 Machine Learning for Computational Design
   - ELL8120 Model Reduction for Control

3. **Industrial Robotics**
   - At most one course in total from the Advanced Control, Reinforcement Learning or Computer Vision basket
   - MEL7202 Mechanical System Design
   - MEL7307 Smart Manufacturing
   - MEL7225 Design Principles for Precision
   - AML7090 Systems Engineering and Design/DDP7121 Introduction to DIY Prototyping
   - BML7730 Industrial Manufacturing for Healthcare
   - AML7830 Digital Twins
   - MEL7103 Operations Planning and Control

- ELL7283 Embedded Systems
- DSL7711 Sensors and Transducers
- DSL7757 Generative AI for Cyber-Physical Systems

4. **Rehabilitation and Medical Robotics**

- At most one course in total from the Advanced Control or Computer Vision basket
- JRL7999 Wearable Robotics
- MEL7225 Design Principles for Precision
- BML8300 Biosensor Technology
- BML7400 Biomedical Instrumentation
- BML7410 Medical Device Design
- BML8800 Healthcare Wearables: Design and Applications
- BML7700 Fundamentals of Biomechanics/MEL7234 Principles of Human Movement/AML7620 Advanced Biomechanics/BML7140 Advanced Neuromechanics
- BML7381 Deep Learning for Medical Image Analysis
- BML7500 Point of Care Medical Diagnostic Devices
- BML7350 Biomedical Signal and Image Processing
- BMP7460 Biomechanics, Rehabilitation Engineering and Haptics Lab
- SIL7020 Accessible Computing & Assistive Technologies

5. **Autonomous and Intelligent Vehicles**

- At most one course in total from the Computer Vision, Advanced Control, Reinforcement Learning, or Autonomy basket.
- JRL8730 Aerial Robotics
- CTL7013 Connected and Autonomous Vehicles
- CTL7003 Introduction to Electric Vehicles/CTL7005 Engineering of Electric Vehicles/ELL7505 Electric Vehicle Systems/CTL7008 Design and Control of EV Powertrain/CTL7032 Vehicle Propulsion and Transmissions/CTL7049 Digital Control Design for EV Applications/CTL7020 Vehicle System dynamics and control

Advanced Control Basket
ELL8101 Learning-Based Control
ELL7125 Control of Robotic Systems
ELL8122 Model Predictive Control
ELL8102 Optimal Control
ELL8100 Nonlinear Control Systems

Reinforcement Learning Basket
AIL7022 Reinforcement Learning

COL7077 Deep Reinforcement Learning
ELL7423 Fundamentals of Reinforcement Learning
AIL8027 Advanced Reinforcement Learning
MEL7118 Reinforcement Learning

Computer Vision Basket
COL8628 Advanced Computer Vision
AIL7031 Deep Learning for Computer Vision

Autonomy Basket
COL7078 Principles of Autonomous Systems
ELL8298 Cognitive Architectures for Agentic AI: Building Intelligent Autonomous Systems